

AI is actually net positive for the environment

Posted by u/Reynvald • 0 points • 10 comments

Take all calculations and sources here with a grain of salt for both sides of the arguments, as such things are generally hard to quantify. I also would be happy to get corrected if I made mistakes or misrepresented some data. And yes, I used various AI tools for research, but manually checked every source that I put in here.

Usual talking points about AI, harming the environment, is:

- Energy consumption
- Carbon footprint and GHG in general
- Water scarcity

1. Energy consumption

As of 2024, [Data centers accounted for about 1.5% of global electricity consumption, with AI accounted for 15% of total data centre energy demand accordingly.](<https://www.scientificamerican.com/article/ai-will-drive-doubling-of-data-center-energy-demand-by-2030/>) Therefore we can say that AI itself is using around 0.225% of global energy reserves.

[Predicted share of energy usage for data centers by 2030 is between 5 and 20%.]

(<https://www.scientificamerican.com/article/ai-will-drive-doubling-of-data-center-energy-demand-by-2030/>) Considering that AI is still on its growth and can take over up to 50% of all data center's resources, in 2030 it can be responsible for 2.5 up to 10% of all energy consumption (20 up to 90 times more, than of now) which is quite radical prediction.

Nevertheless, as of right now, [ML-related technologies is able to provide 15% improvement in grid efficiency and 10–20% increase in battery storage efficiency](<https://www.mdpi.com/2073-4433/15/10/1250>) and [20–30% relative efficiency gains in cell and module R&D.]

(<https://journals.sagepub.com/doi/full/10.1177/0958305X241256293>)

Same magnitude of efficiency gains is also the case for all clean and non-clean energy sources, by forecasting the weather and autoadjusting solar panels, micromanaging power grids and plants, predicting deposits of fossil energy sources and so on.

Safe to say, that estimated energy gain overall will equal to or most likely surpass even the most pessimistic prognosis of 10% energy consumption from AI alone by 2030.

2. Carbon footprint and GHG in general

According to [ICEF report from November 2024](https://www.icef.go.jp/wp-content/themes/icef/pdf/2024/roadmap/15_ICEF2.0%20GHG%20Emissions%20from%20AI_stand%20alone.pdf), (This link will download PDF file!) AI's total GHG emissions are estimated at 100–300 million tonnes CO₂, or roughly 0.2–0.6% of global emissions. With that, operational emissions are around 0.05% while manufacturing servers, chips, facilities, model trainings and life-cycle impacts make up the remainder.

At the same time AI can [reduce global GHG emissions by 5–10% by 2030] (<https://thesustainableagency.com/blog/environmental-impact-of-generative-ai/>), via optimized grids, predictive maintenance, and smart agriculture and, additionally, [cuts of up to 5.3 gigatons CO₂ (another 5–10% of current emissions)](<https://www.sciencedirect.com/science/article/pii/S277242712200122X>) - through applications in transport, buildings, and supply chains.

[One specific research](<https://www.nature.com/articles/s41598-025-97110-3>) (from month ago) from China indicates, that correlation between % of AI adoption and % of reducing carbon footprint (1% and 0.0395% accordingly) is quite sustainable and universal across the industries.

3. Water scarcity

There is not much fresh unbiased data and peer-reviewed papers on AI water consumption. [Apparently in US AI is responsible for 0.5–0.7% of total annual water withdrawal](<https://oecd.ai/en/work/how-much-water-does-ai-consume>). If source took a data of water consumptions by data centers in general (it most likely the case), then actual numbers will be a 15% of 0.5–0.7%, which is 0.075–0.105% accordingly.

Considering that most of the world AI infrastructure is located in US and China, safe to say, that for the rest of the world this percentages is significantly smaller.

The real concern, however, is the water pollution (which is still extremely small, compared to the heavy and construction industries) and separate cases of mismanagement from the corporations. [Quote: "Google's planned data centre in Uruguay, which recently suffered its worst drought in 74 years, would require 7.6 million litres per day, sparking widespread protest."](https://sdgs.un.org/sites/default/files/2024-05/Gupta%2C%20et%20al._AIs%20excessive%20water%20consumption.pdf) (This link will download PDF file!)

Now to a good news:

[AI irrigation can reduce water usage by 30–50% while increasing yields by 20–30% (which is 5–8% savings of global agricultural withdrawals if deployed worldwide).] (<https://www.sciencedirect.com/science/article/pii/S2772375525002151>)

[AI acoustic and pressure-based leak detection is already working and have 80–97% accuracy, cutting non-revenue water losses by 20–40%. Given that networks lose ~30% of supply globally (the most distant and arid places usually suffer the most), AI is saving 6–12% of treated water.] (https://www.waterrf.org/serve-file/2024_IWS-Challenge-Solution_Case-Western-CWD.pdf) (This link will download PDF file!)

Same goes for demand forecasting, pump optimization, water quality assessment and many other projects, totaling up to 12% of the saved fresh water worldwide (if implemented worldwide as well). Some of this solutions is already implemented and working, although mostly in the most water hungry areas, like parts of Africa, China and India.

There is crucial to point out, that most of the water scarcity-related suffering is occurring far from data centers and their water sources. And this problem is a logistical one (how to transport the water to the arid areas), not the problem of sheer amount of fresh water world supplies.

Fun facts, regarding the general misconception that AI consume literally bottles of water per query:

.1. [The amount of water, that ChatGPT needs to consume is around 500ml of water *for 10-50 queries*

This means that each query is about $500/30=17\text{ml}$.]

(<https://miljamoss.neocities.org/Articles/LLMWaterAndEnergyUse>)

- The amount of water required to produce an 8oz steak is 3,217,000 ml. So you would need to make 189,000 queries to equal the water cost of a steak dinner.

- Average shower uses about 8000ml of water per minute. So you'd have to make 470 queries to use the amount of water you spend if you're in the shower for one extra minute.

- Finally, flushing the toilet uses 6000ml. So if you pee one extra time per day that's about 350 queries.

.2. [Humans themselves is far more environmentally impactful, compared to AI, when performing the same tasks. Hundreds times so, even.](<https://www.nature.com/articles/s41598-024-54271-x>)

I want to highlight, that AI still have an impact on environment and it's a right thing to strife for reducing the environmental impact in any area. But I believe that misinformation, toxicity and alarmism eventually will harm the both sides of this debates.

Comments

u/Starshot84 • 1 points • 2025-05-30 16:27 UTC

Your data is out of date. AI energy cost is going up on a monthly if not weekly basis, increasingly.

u/Lopsided-Yak9033 • 2 points • 2025-06-06 18:44 UTC

So far since finding myself in this sub, it's pretty evident that the word fact is heavily abused.

Most posts seem to be an opinion delivered wrapped in a very pedantic fact, a dubious or outdated study that doesn't really assert the claim as fact, or an outright side stepping of the reality they're arguing.

Even if these studies were currently up to date research and 100% correct, at most you could factually say is "some predictions show AI could yield a net positive result for the environment by these measures." The actual fact is - we cannot state with certainty how AI will affect the environment in the future because it is still an emerging field.

u/Nexinex782951 • 1 points • 2025-05-25 19:33 UTC

Current greenhouse gas trajectories indicate that what we need is degrowth--a slight decrease in overall energy consumption--if we want to stay on target for avoiding some real catastrophic things anytime soon. This is quite simply not the time to be increasing energy production, as we cannot do it green enough to avoid disaster yet. Now, is any country going to do degrowth? No. Is it still possible to make it worse by scaling up energy costs? Already is happening. Your argument purports efficiency gains. The problem is having an llm on every single Google search doesn't gain in efficiency. Research? Definitely a worthwhile use. But it requires you lump everything together to make this argument. There's a huge difference between so called generative AI and whatever the term for "functional" AI is.

u/SIUonCrack • 1 points • 2025-05-13 03:02 UTC

5-20% efficiency improvements don't matter when AI is projected to be >50% new energy demand growth.

u/irritated_socialist • 0 points • 2025-05-11 19:54 UTC

Right now, there is a child with the potential to crack the fusion problem once and for all. Without state intervention, ChatGPT and its snowclones will prevent her mind from even approaching its potential.

u/Reynvald • 1 points • 2025-05-11 21:18 UTC

Right now there is a child somewhere in a rural Africa, with a potential to incorporate gravity into Standard Model. Without ChatGPT and its snowclones she might never be taught and challenged enough to reveal her full potential.

<https://www.fepbl.com/index.php/ijarss/article/view/725>

u/Free_Tumbleweed_860 • 2 points • 2025-05-11 17:58 UTC

Does it really matter? None of these companies give a shit.

u/Reynvald • 2 points • 2025-05-11 18:14 UTC

Most of the large companies have initiatives on this matter. And this companies is objectively greener in proportion, even if they consume more resources than 10-20 years ago. Of course main concern is profit (and it makes all the sense). But with enough pressure from state and people it's pretty realistic to force companies to give a shit. Many big business in some European countries is already have zero or near zero carbon footprint, for example.

u/Free_Tumbleweed_860 • 1 points • 2025-05-11 18:16 UTC

They won't do anything unless forced to because they do not care.

Profits > environment

u/Reynvald • 2 points • 2025-05-11 18:21 UTC

Yes, it's exactly what I said. Companies rarely created with a first goal to save the planet. So they need a push on that matter. And I fully support those who pushing them.

u/Complex_Package_2394 • 1 points • 2025-05-11 17:03 UTC

"AI big energy consumption bad" is an old tale that gets regurgitated every time a new tech is about to change things.

When you had the first cars in public, the discussion was about whether it's worth producing oil to power them in contrast to horses just eating grass. Part of that was also "people are going to use it for leisure, so it's wasting resources" but of course energy consumption/upkeep per mile was way lower than with horses after some technical developments.

As the price per mile declined, cars and trucks got used for more and more things, making transportation in general way more efficient. Car usage led to a significant increase in energy consumption, just like AI will, but that's because the extra efficiency was worth it. Especially the saved man hours made it worth it.

AI will, for the resources invested, produce way more value than if we choose to not use it. A better value to resource ratio will make it more viable, enlarging its usage and demand, increasing resource consumption and further increasing value.

Good read to that is the Jevons Paradox (https://en.m.wikipedia.org/wiki/Jevons_paradox), the same thing happened with steam engines and coal consumption in the UK.

People using AI instead of googling sure is a bit wasteful, but most of the energy demand will not be produced by people hitting up ChatGPT with bullshit questions but by business and governments actually using AI to make their thing more productive.

u/soowhatchathink • 8 points • 2025-05-11 16:38 UTC

It looks like you're conflating use-case specific AI used explicitly for environmental purposes and generic LLMs that the general public are using.

Adding an LLM response to every Google search is not beneficial for the environment and neither is the way we use ChatGPT.

u/Reynvald • 0 points • 2025-05-11 16:58 UTC

Thanks for reply. I would argue with it on three occasions.

1. You can't really create a technology and then limit it for narrow window of cases. It might be good for some things, but it just almost never works. So we have to take it as a given. Besides, each query is still have a complete negligible impact, compared with everyday things that we taking for granted.
2. Using LLM for everyday life (considering that it evidently save people's time), might also have its environmental advantages. According to Nature journal (there is a link in my post) LLM much less wasteful than humans at the respective tasks.
3. This exact general public usage is precisely what sponsoring development and implementation use-case specific AI.

u/AutoModerator • 2 points • 2025-05-11 12:21 UTC

Backup in case something happens to the post:

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